

Aims and contributions

Early detection of breast cancer is key to survival. The local Hölder exponent distribution in a mammogram enables to quantify breast tissue disruption, and hence to assess breast cancer risk. This work proposes a systematic study of the detectability of disrupted tissues embedded inside fatty vs. dense microenvironments leveraging simulated piecewise homogeneous fractal textures modeling breast tissues.

Contributions:

- Novel *filtered* fractional Brownian field (*filtered fBf*) model for stationary isotropic fractal textures.
- Simulations on synthetic piecewise homogeneous *filtered fBf* and fractional Gaussian field (*fGf*) textures: segmentation of a patch.
- Quantification of the detectability of simulated disrupted tissues $H_p = 0.5$ in simulated fatty $H_b = 0.3$ vs. dense $H_b = 0.65$ tissues.

Outcomes:

- High detection performance for large patches of disrupted tissues in **fatty** environments, with a drop in accuracy for small patches.
- For **dense** environments detection performance are good and decrease slowly with the patch size.

Mammogram provided by **CompUMaine** from (Gerasimova-Chechkina et al., 2021, *Front. Physiol.*)



Fractal texture models

Fractional Brownian field (*fBf*)

$$B_H(\underline{x}) = \int_{\mathbb{R}^2} \frac{e^{-i\underline{x} \cdot \underline{\omega}} - 1}{\|\underline{\omega}\|^{H+1}} d\tilde{W}(\underline{\omega})$$

(Mandelbrot & Van Ness, 1968, *SIAM Rev.*)

Stationary self-similar fields

Fractional Gaussian field (*fGf*)

$$G_H(\underline{x}) = B_H(\underline{x} + \underline{e}_1) + B_H(\underline{x} + \underline{e}_2) - 2B_H(\underline{x})$$

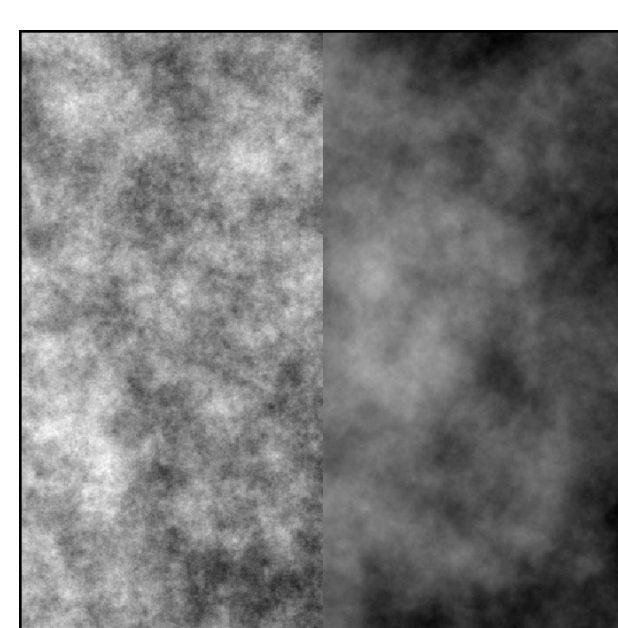
(Pascal et al., 2021, *Appl. Comput. Harm. Anal.*)

Filtered fractional Brownian field

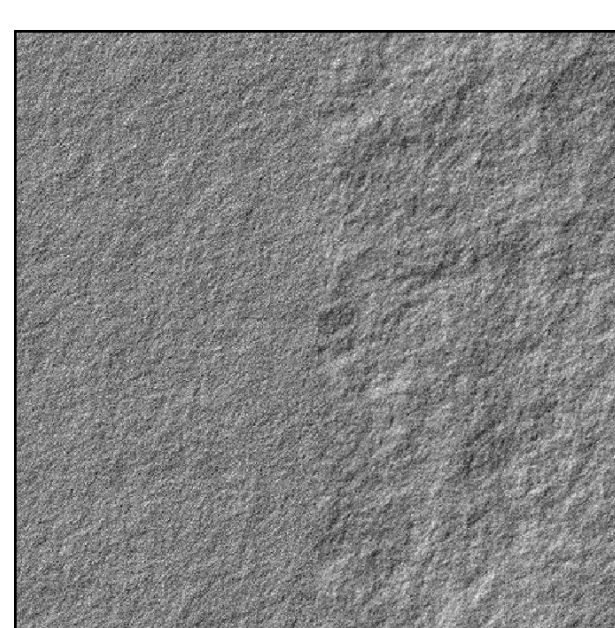
$$C_H(\underline{x}) = \langle B_H, \underline{u}_{\underline{x}} \rangle, \text{ u high-pass filter}$$

$\langle \cdot, \cdot \rangle$ scalar product in $L^2(\mathbb{R}^2)$

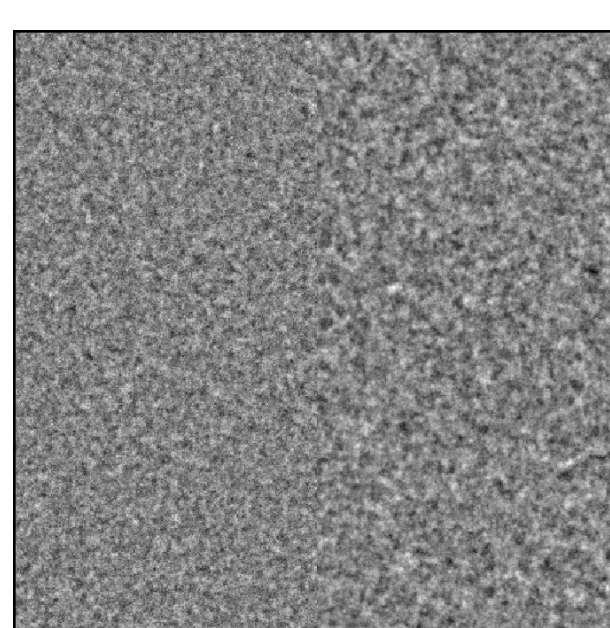
Filter u from (Biermé et al., 2024, *Preprint*)



fBf B_H



fGf G_H



Filtered fBf C_H

Piecewise textures

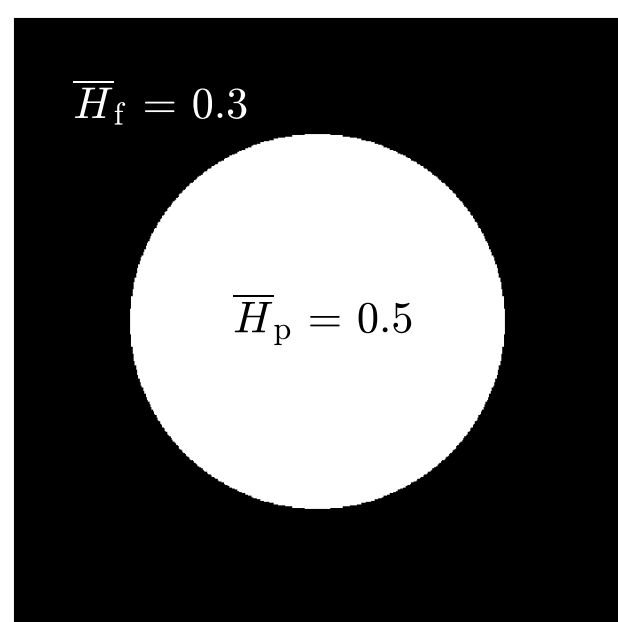
Goal: Model breast local microenvironment.

Background: healthy microenvironment

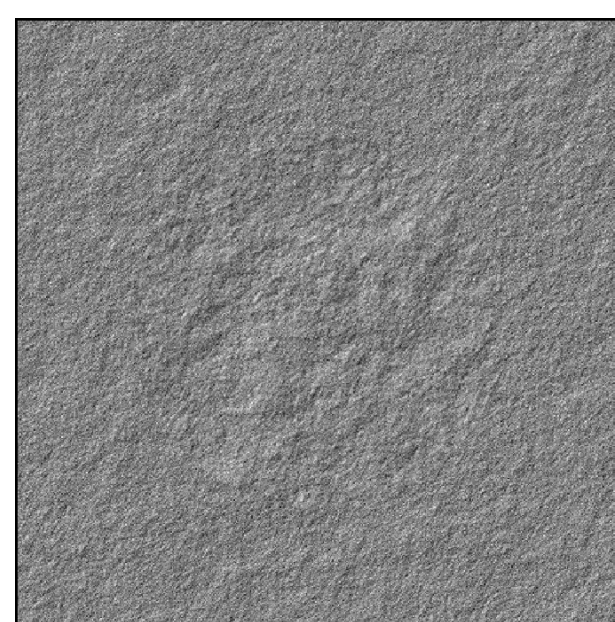
- fatty tissues: $H_b = 0.3$ (anticorrelated)
- dense tissues: $H_b = 0.65$ (correlated)

Patch: disrupted tissues

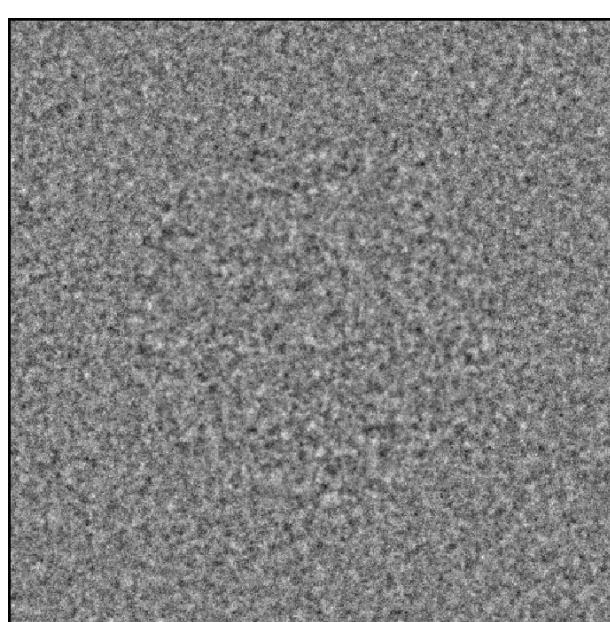
- $H_p = 0.5$ (uncorrelated)



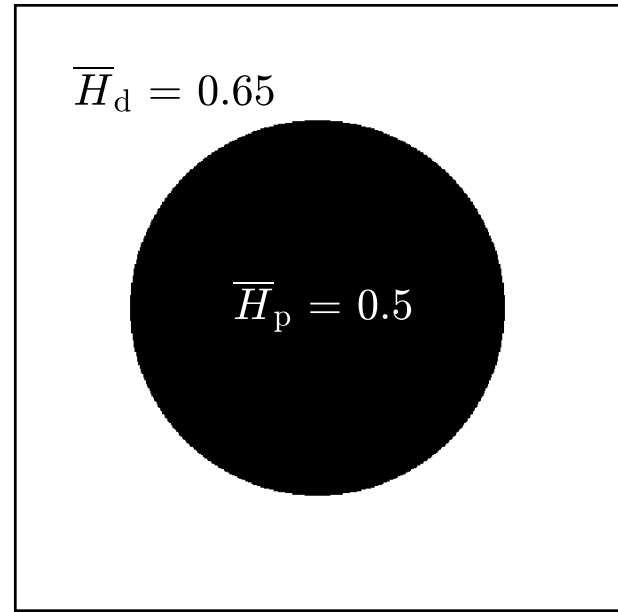
Fatty background



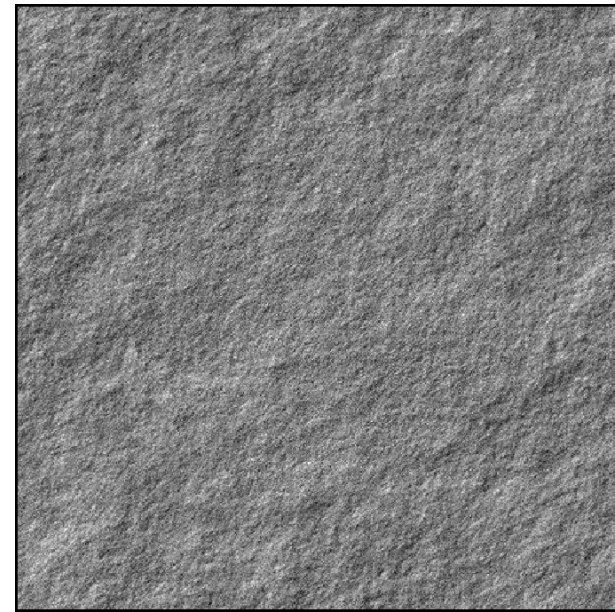
fGf



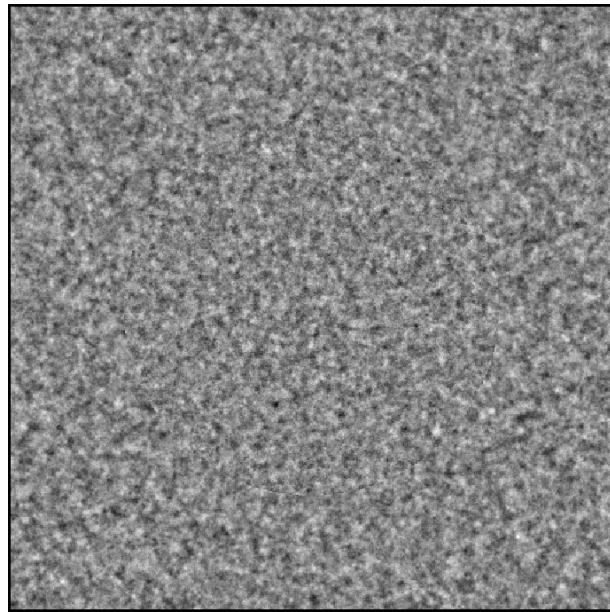
Filtered fBf



Dense background



fGf



Filtered fBf

Examples:

- images of 512×512 pixels;
- circular central patch: 30% of pixels;
- two configurations: two background types.

Hölder exponent-based texture segmentation

Local Hölder exponent: for a field $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\underline{x}_0 \in \mathbb{R}^2$, $h(\underline{x}_0)$ defined as the largest exponent $\alpha > 0$ such that there exists a constant χ and a polynomial $\mathcal{P}_{\underline{x}_0}$ of degree lower than α such that for $\underline{x}$ in a neighborhood of $\underline{x}_0$: $|F(\underline{x}) - \mathcal{P}_{\underline{x}_0}(\underline{x})| \leq \chi \|\underline{x} - \underline{x}_0\|^\alpha$. For B_H , G_H and C_H : $\forall \underline{x} \in \mathbb{R}^2$, $h(\underline{x}) = H$.

Multiscale analysis and wavelet leaders: (Mallat, 1999, *Elsevier*) scaling function ϕ , mother wavelet $\psi \Rightarrow \mathcal{Y}_F^{(m)}(j, \underline{k}) = 2^{-j} \langle F, \psi_{j, \underline{k}}^{(m)} \rangle$; *leaders*: $\mathcal{L}_{j, \underline{k}} = \sup\{|2^j \mathcal{Y}_{j', \underline{k}'}^{(m)}|, \lambda_{j', \underline{k}'} \subset 3\lambda_{j, \underline{k}}, m = 1, 2, 3\}$

$$\mathcal{L}_{j, \underline{k}} \simeq \eta(\underline{x}) 2^{j h(\underline{x})} \quad \text{as } 2^j \rightarrow 0 \quad (\text{Jaffard, 2004, Proc. Symp. Pure Math.})$$

$\Rightarrow$ Linear Regression on $\log_2 \mathcal{L}_{j, \underline{k}}$: local estimate $\hat{h}^{\text{LR}}(\underline{x})$.

Threshold Rudin-Osher-Fatemi estimator: 2D discrete gradient operator $\mathbf{D}$

$$\hat{\mathbf{h}}^{\text{ROF}} = \underset{\mathbf{h}}{\text{argmin}} \|\mathbf{h} - \hat{\mathbf{h}}^{\text{LR}}\|_2^2 + \lambda \|\mathbf{D}\mathbf{h}\|_{2,1} \text{ \& iterative thresholding } \Rightarrow \mathcal{T}\hat{\mathbf{h}}^{\text{ROF}}.$$

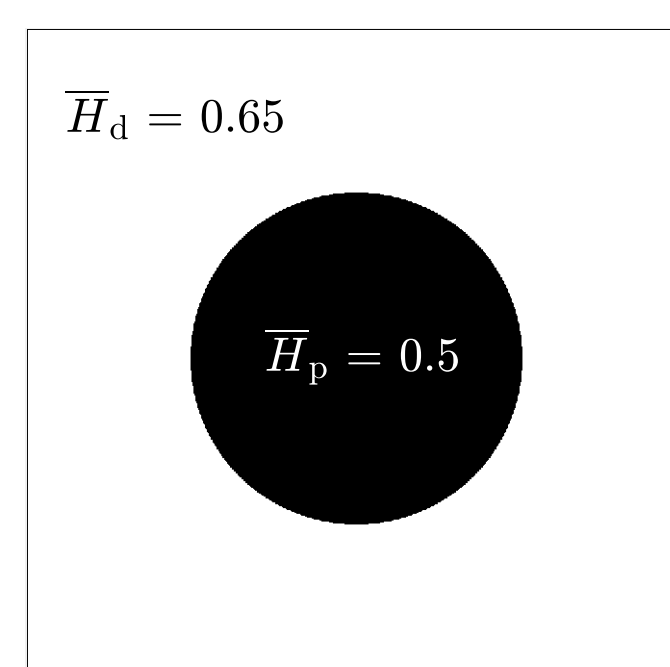
(Nelson et al., 2016, *IEEE Trans. Image Process.*; B. Pascal et al., 2018, *ICASSP*; Cai et al., 2013, *SIAM J. Imaging Sci.*; Pascal et al., 2021, *Appl. Comput. Harm. Anal.*)

Stein-based automated parameter tuning: adapted Generalized Stein Unbiased Risk Estimate

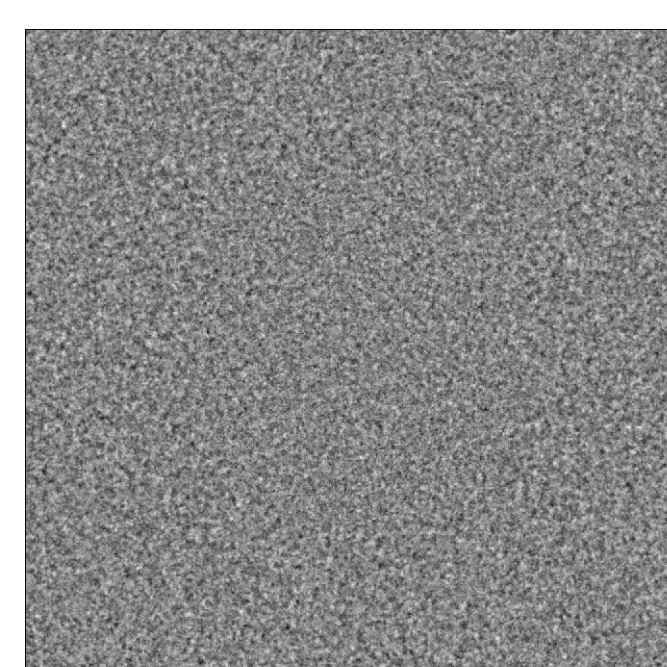
$$\text{GSURE}(\lambda) = \left\| \hat{\mathbf{h}}^{\text{ROF}} - \hat{\mathbf{h}}^{\text{LR}} \right\|^2 + 2\text{Tr}(\mathcal{S}\mathbf{J}) - \text{Tr}(\mathcal{S}) \text{ not explicitly depending on } \bar{h}$$

$\mathbf{J}$: Jacobian matrix of $\hat{\mathbf{h}}^{\text{ROF}}$ w.r.t. $\hat{\mathbf{h}}^{\text{LR}}$; $\mathcal{S}$: empirical covariance of Gaussian noise corrupting $\hat{\mathbf{h}}^{\text{LR}}$.

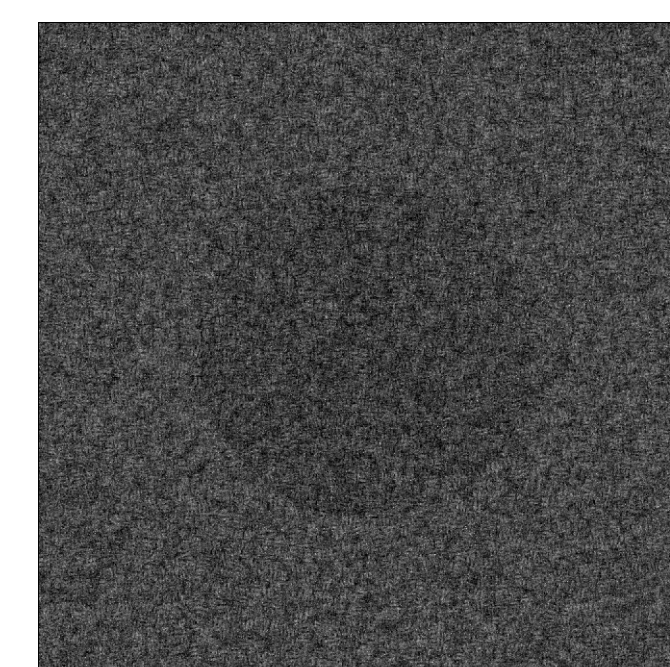
Optimal λ^* : minimization of $\text{GSURE}(\lambda)$ with a BFGS scheme. (Pascal et al., 2020, *Ann. Telecommun.*)



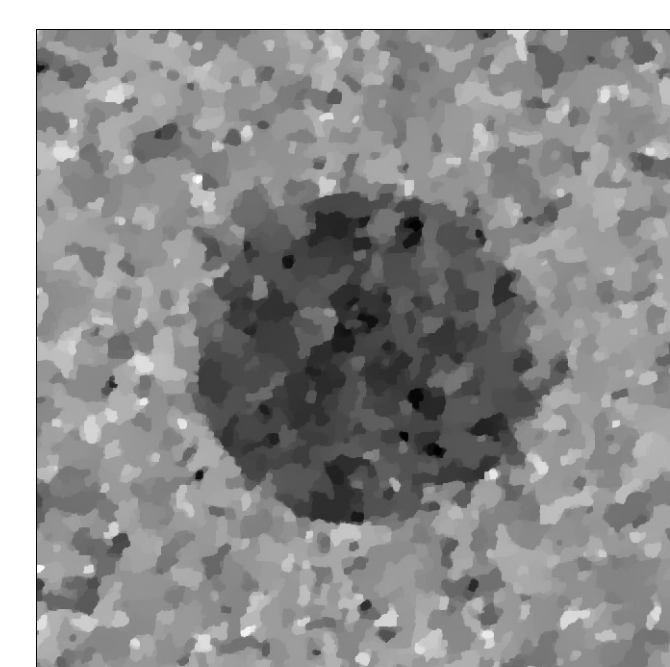
True $\bar{h}$



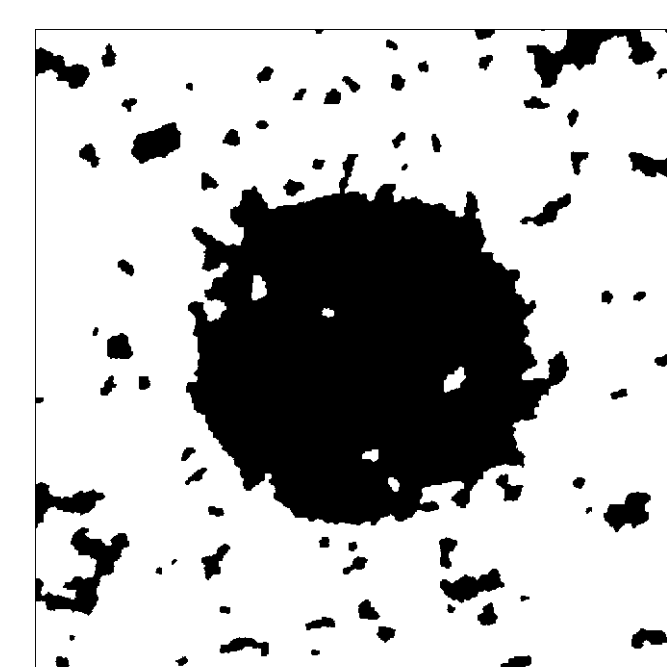
Piecewise texture



Lin. reg. $\hat{h}^{\text{LR}}$



ROF $\hat{h}^{\text{ROF}}$



T-ROF $\mathcal{T}\hat{h}^{\text{ROF}}$

Numerical experiments: detectability of disrupted tissues

Detection performance criteria: F-score defined as $F_1^{-1} = \text{precision}^{-1} + \text{recall}^{-1}$

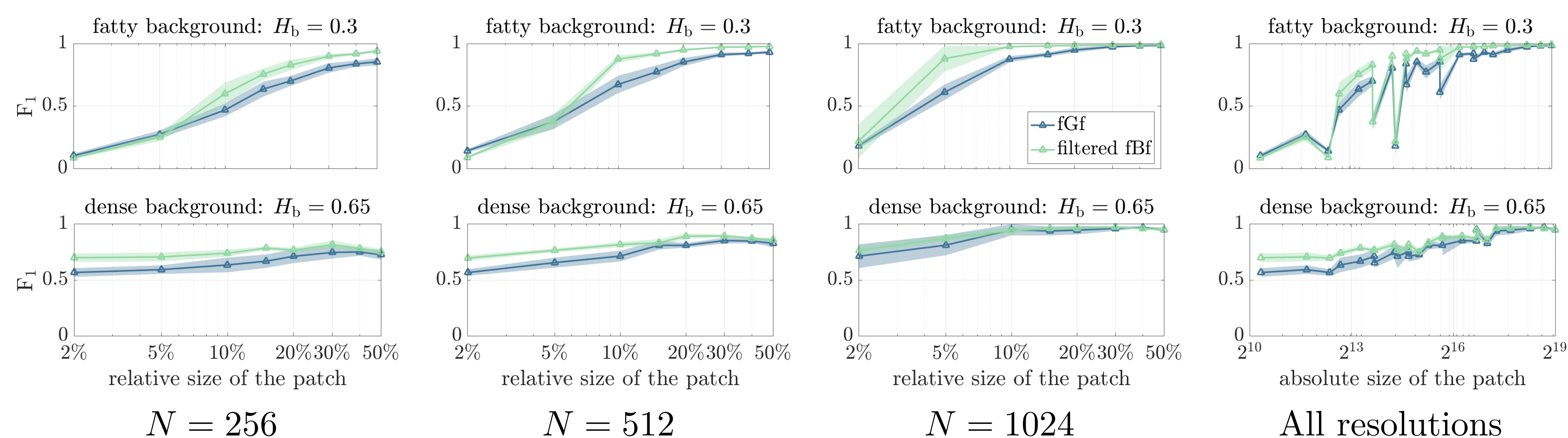
- precision: proportion of pixels segmented in the central patch indeed belonging to it;
- recall: proportion of pixels originally belonging to the central patch and correctly segmented.

$\Rightarrow$ The larger $F_1 \in [0, 1]$ the better the segmentation in terms of both errors of types I and II.

Two configurations: patch of disrupted tissues embedded in *fatty* vs. *dense* background; for each

- eight *relative* patch sizes $\{2\%, 5\%, 10\%, 15\%, 20\%, 30\%, 40\%, 50\%\}$,
- three image resolutions $N \in \{256, 512, 1024\}$,
- two synthetic texture models: *fGf* vs. *filtered fBf*,

$\Rightarrow$ F_1 score average and 95% confidence regions computed on 10 texture realizations.



Perspectives

- Detectability of disrupted tissues in *anisotropic* textures (Richard & Biermé, 2010, *J. Math. Imaging Vis.*),
- Confidence level on Hölder exponent-based risk cancer assessment on real datasets, **VinDr-Mammo**.